

Lucca da Paz

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PROFILE

Mathematics graduate specialized in Data Engineering, with hands-on experience building streaming pipelines (Kafka, Flink), Lakehouse architectures (Iceberg, Paimon, MinIO), orchestrated ETL workflows (Airflow, dbt), and analytical data modeling. Additional background in Machine Learning and Applied AI (TensorFlow, RAG systems, custom neural networks).

PROJECTS

Real-Time Streaming Pipeline for Transaction Risk Analysis Python, Flink, Kafka, Paimon *Data Engineering Project | 2026*

- Built a real-time streaming pipeline with Apache Flink 2.2 and Apache Kafka for transaction processing and risk scoring, orchestrated via Docker Compose.
- Designed a Lakehouse storage layer with Apache Paimon on S3-compatible storage (MinIO) for downstream analytical workloads.
- Instrumented jobmanager/taskmanager with Prometheus metrics for pipeline monitoring and observability.
- Integrated local LLM inference (Ollama) into the pipeline for enrichment tasks.

Lakehouse & RAG Recommendation System Python, Spark, Airflow, Iceberg *Data Engineering Project | 2026*

- Built an end-to-end Lakehouse architecture with Apache Iceberg (Silver/Gold layers), MinIO, and a REST catalog backed by Postgres, orchestrated with Apache Airflow.
- Developed a recommendation model (Spark MLlib ALS) with temporal decay and blockbuster-penalization logic, enriched via external API ingestion (TMDb).
- Generated embeddings with Ollama (nomic-embed-text) stored in Postgres via pgvector, powering a RAG chatbot served through a Streamlit application.

Analytics Engineering Pipeline (Airbnb) dbt, Athena, Evidence *Data Engineering Project | 2026*

- Built a dbt-based transformation pipeline on top of AWS Athena for Airbnb listings data, structuring raw data into analytics-ready models.
- Developed automated analytical reports with Evidence, connected directly to the Athena data source.

Machine Learning & Neuro-Fuzzy Modeling Projects Python, TensorFlow, Keras *Machine Learning Projects | 2026*

- Implemented a custom Sequence-to-Sequence model with Bahdanau Attention for character-level date normalization into ISO 8601 format.
- Developed an ANFIS (Adaptive Neuro-Fuzzy Inference System) model based on the Takagi-Sugeno inference system, with Wang-Mendel rule extraction.

TECHNICAL SKILLS

Data Engineering: Apache Spark, Kafka, Flink, Airflow, dbt, Iceberg, Paimon, MinIO, DuckDB, Athena

Databases: PostgreSQL, pgvector, SQL

Machine Learning: TensorFlow, Keras, Scikit-Learn, Neural Networks, Attention Mechanisms, NLP, RAG

Infrastructure & Tools: Linux, Docker, Docker Compose, Prometheus, Git

Programming: Python, Julia, SQL, Bash, LaTeX

EDUCATION

University of Campinas (UNICAMP)

August 2024 – September 2026

Campinas, Brazil *M.Sc. in Applied Mathematics*

Federal University of São Carlos (UFSCar)

March 2020 – August 2024

São Carlos, Brazil *B.Sc. in Mathematics*

LANGUAGES

English: Advanced (Professional Working Proficiency)

German: Basic